Borg: Open-Source Tile-Based GPU with Silicon Tapeout

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Motivation

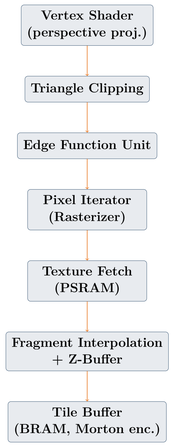
GPU architectures are ubiquitous in rendering research, yet every commercially deployed GPU keeps its microarchitecture, instruction set, and silicon layout proprietary. This impedes **reproducibility**, **teaching**, and **architectural experimentation**.

Borg is a fully open response: a Tile-Based Deferred Rendering (TBDR) GPU implemented in Chisel, synthesized with an open toolchain, and submitted to physical silicon — twice.

Key claim: *To our knowledge, Borg is the only open-source GPU for which a physical ASIC tapeout has been completed.*

TBDR Architecture

Borg follows the Tile-Based Deferred Rendering paradigm used by **PowerVR / Imagination Technologies**, **Apple GPU**, **ARM Mali**, and **Vivante**:



Geometry is processed tile-by-tile; only *visible* fragments are shaded, minimising external memory bandwidth — critical for embedded FPGAs with limited PSRAM bandwidth.

Completed Pipeline (Steps 1–20)

All 20 steps completed in **28 days** (19 Mar – 14 Apr 2026):

	Component	Date
✓	Hardware Edge Function Unit	2026-03-19
✓	Multiple Triangles (Firmware)	2026-03-20
✓	32-bit RISC-V Expansion	2026-03-23
✓	Larger Register File + IMEM	2026-03-23
✓	Perspective Projection (HW)	2026-03-24
✓	Hardware RCP Unit	2026-03-24
✓	Triangle Clipping (HW)	2026-03-27
✓	Hardware Fragment Interpolation	2026-03-27
✓	Arcilator + Verilator simulation	2026-03-29
✓	Pixel Iterator (HW Rasterizer)	2026-04-05
✓	Hardware Z-Buffer Unit	2026-04-07
✓	Command FIFO	2026-04-08
✓	SystemRDL Register Description	2026-04-08
✓	Interactive Viewer	2026-04-10
✓	Texture Fetch Unit	2026-04-13
✓	LUT Recovery	2026-04-13
✓	SoC Project Restructure	2026-04-13
✓	Shared Memory Controller	2026-04-14
✓	IO Bundle Refactor	2026-04-14

Current Demo

Running on **pico-ice** (iCE40 UP5K, 5280 LCs):

- 12 textured triangles, 32×32 framebuffer
- Full pipeline: vertex shader → texture fetch → Z-buffer → output
- CPU-orchestrated (GPU autonomy = Phase 2, in progress)

Bandwidth note: Current bottleneck is shared CPU/GPU PSRAM access. GPU DMA engine (Step 22) will eliminate CPU involvement in rendering.

FPGA Resource Budget (iCE40 UP5K)

Resource	Used	Available
Logic Cells	∼5 100	5 280
BRAMs (4 kb)	∼10	30
DSPs	0	8

Operating within a $20 FPGA forces disciplined area optimisation: nibble-serial FMAs, shared memory controllers, and PeakRDL-generated register files contribute to fitting a full GPU pipeline in 5 280 logic cells.

ASIC Tapeouts

Run	Process	Content	Date
TTIHP26a	IHP SG13G2 130 nm	FP adder peripheral	Mar 2026
TTSKY26a	SkyWater SKY130	Triangle rasteriser	May 2026
GF180	GlobalFoundries 180 nm	Diffuse shader ASIC	2022

All tapeouts via the **Tiny Tapeout** open-access silicon programme and the **Google Open-Source Silicon Programme**. Chips for TTIHP26a expected July 2026.

Open-Source Toolchain

HDL	Chisel (Scala) → FIRRTL → Verilog
Synthesis	Yosys
P&R FPGA	nextpnr-ice40
P&R ASIC	OpenROAD / OpenLane
Simulation	Verilator, cocotb, Arcilator
Reg. map	PeakRDL-chisel

Comparison with Related Open-Source GPUs

	Borg	Nyuzi	Vortex
Architecture	TBDR	GPGPU	RISC-V GPGPU
ISA	Custom	Custom	RV32
Tapeout	✓	—	—
Open toolchain	✓	partial	partial
FPGA target	iCE40	Arria 5	Xilinx
Focus	Graphics	Compute	Compute

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Future Work

- [wip] **GPU autonomy** (Steps 21–26): DMA engine, write port, integrated vertex sequencer, real-time VGA output
- ULX3S / ECP5 port for higher resolution
- ISA extensions for BVH ray traversal (path tracing on Borg)
- Integration with Gonzales path tracer: first fully open-source hardware path tracer

References

[1] Woop et al. *RPU: A Programmable Ray Processing Unit*, SIGGRAPH 2005.
Borg GPU book: https://gonsolo.github.io/Borg
Source: https://github.com/gonsolo/Borg

LaTeX TikZposter